POST-INTERVIEW TECHNICAL ASSESSMENT REPORT

Mechanistic Digital Twin & Stochastic Risk-of-Collapse Simulation in Biologics Lyophilization under Edge-Vial Heat Transfer Variability

Case Study: From Real-World Problems to Process Simulation | **Applied Role:** Data & Process Simulation Engineering (PSE)

Target Facility: PT Kalbio Global Medika (KGM) — Cikarang	**Focus Domain:** Sterile Parenteral Biomanufacturing	**Candidate:** Ahmad Fathin Naufal
Regulatory Framework: ICH Q8(R2) Quality by Design (QbD)	**Core Methodology:** 1D Pikal ODE + 1,000 Monte Carlo	**Track:** Process Simulation Engineer Intern

1. Problem Exploration & Industrial Background

1.1 Industrial Context at Kalbe Biologics: PT Kalbio Global Medika (KGM), a biopharmaceutical manufacturing subsidiary of PT Kalbe Farma Tbk located in Cikarang, West Java, represents Indonesia's premier advanced biomanufacturing facility. Operating under strict current Good Manufacturing Practice (cGMP) aligned with BPOM, EMA, and PIC/s standards, KGM manufactures high-value therapeutic proteins, including recombinant human erythropoietin (EPO), granulocyte colony-stimulating factor (G-CSF), and monoclonal antibodies (mAbs). Because therapeutic macromolecules exhibit severe conformational instability and hydrolytic degradation in aqueous solution, **lyophilization (freeze-drying)** is the indispensable unit operation used to confer multi-year shelf-life stability.

1.2 The Industrial Dilemma & Economic Stakes: Primary drying is the most critical and energy-intensive stage of freeze-drying, typically requiring 48 to 72 hours per commercial batch. During this phase, shelf heat drives sublimative ice removal under deep vacuum. If the product interface temperature (T_p) exceeds the formulation's **critical collapse temperature ($T_c \approx$ -29.0°C)**, micro-collapse occurs: the amorphous glassy cake softens, pore channels seal, and cake structure irreversibly deteriorates. The financial consequence of a single collapsed batch ranges from **IDR 500 million to over 2 billion** in lost Active Pharmaceutical Ingredient (API), steriles cleanroom occupation, and regulatory deviation investigations. Consequently, plant operators historically adopt excessively conservative shelf temperatures (e.g., -32°C), which penalizes annual manufacturing throughput by 10 to 18 hours per cycle.

1.3 Physical Root Cause of Uncertainty — The Edge-Vial Effect: In industrial freeze-dryers, vial populations do not experience homogeneous heat transfer. Vials located at the perimeter (edges and corners of the tray) receive significant supplementary radiant energy from the warmer chamber walls and doors. As documented by Rambhatla and Pikal (2003), perimeter vials exhibit an apparent vial heat transfer coefficient (K_v) that is **15% to 25% higher** than center vials. This positional stochasticity renders deterministic single-vial models dangerously inadequate for commercial scale-up.

Core Engineering Question:
'What multidimensional operating window ($T_{shelf} \times P_{chamber}$) maximizes cycle-time reduction during primary drying while ensuring that the batch-wide probability of cake collapse $P(T_p > T_c)$ remains strictly below 0.1% under real-world stochastic thermal heterogeneity?'

1.4 System Boundaries & Control Parameters: The freeze-dryer is treated as a coupled thermodynamic control volume. The two manipulated variables (Critical Process Parameters — CPPs) are shelf temperature setpoint T_{shelf} and chamber total pressure P_c (controlled via sterile nitrogen bleed). The primary response variable (Critical Quality Attribute — CQA) is product interface temperature $T_p(t)$ relative to collapse threshold T_c, with secondary CQA being primary drying completion time t_{dry}.

2. Mechanistic Modeling Formulation & Deterministic Verification

2.1 First-Principles System Conceptualization: To establish a predictive Digital Twin, we implement the 1D quasi-steady-state moving-boundary formulation originated by Pikal et al. and formalized by Fissore and Barresi (2011). The formulation rigorously couples heat conduction, gas rarefaction, radiation, and Darcy-Knudsen vapor diffusion:

(Eq. 1) Heat Influx:
$q = A_v \cdot K_v \cdot (T_{shelf} - T_p)$

(Eq. 2) Sublimation Mass Flux:
$dm/dt = [A_p \cdot (P_{ice}(T_p) - P_c)] / R_p(L)$

(Eq. 3) Clausius-Clapeyron Vapor Pressure:
$P_{ice}(T) = exp(28.8926 - 6140.4 / T)$ [Pa]

(Eq. 4) Dynamic Cake Resistance:
$R_p(L) = R_0 + A_1 \cdot L$ [Pa·s·m²/kg]

(Eq. 5) Interfacial Energy Balance:
$K_v(T_{shelf} - T_p) = [\Delta H_{sub} \cdot (P_{ice}(T_p) - P_c)] / R_p(L)$

(Eq. 6) Sublimation Front Progression (ODE):
$dL/dt = (dm/dt) / [\rho_{ice} \cdot A_p \cdot (1 - \varepsilon)]$

2.2 Empirical Grounding & Parameter Registry: Physical constants are calibrated from published literature for 5% w/v sucrose/mannitol protein formulations in standard Schott 10R tubing vials (outer diameter 24.0 mm, fill volume 3.0 mL, initial cake height L_{total} = 8.19 mm):

Parameter Description	Symbol	Calibrated Value [SI]	Literature Source
Latent Heat of Sublimation	ΔH_{sub}	2.839×10^6 J/kg	CRC Handbook of Chemistry & Physics
Ice Density & Porosity	ρ_{ice}, ε	917 kg/m³, 0.92	Pikal et al. (2004)
Mean Heat Transfer Coeff.	$K_{v,mean} \pm \sigma$	25.0 ± 3.5 W/(m²·K)	Rambhatla & Pikal (2003)
Initial Cake Resistance	R_0	3.60×10^5 Pa·s·m²/kg	Fissore, Pisano, Barresi (2011)
Resistance Slope Factor	A_1	4.20×10^8 Pa·s·m/kg	Pisano, Barresi, & Fissore (2011)
Critical Collapse Temp.	T_c	-29.0 ± 0.8 °C	Freeze-Dry Microscopy benchmark

2.3 Numerical Algorithm: At each 60-second time increment, Eq. 5 is solved implicitly for product interface temperature T_p using Brent's method (`scipy.optimize.brentq`), achieving residual energy convergence < 10^{-6} K. Sublimation flux dm/dt is evaluated, and the dried layer boundary L is advanced via explicit Euler integration until complete cake drying ($L = L_{total}$).

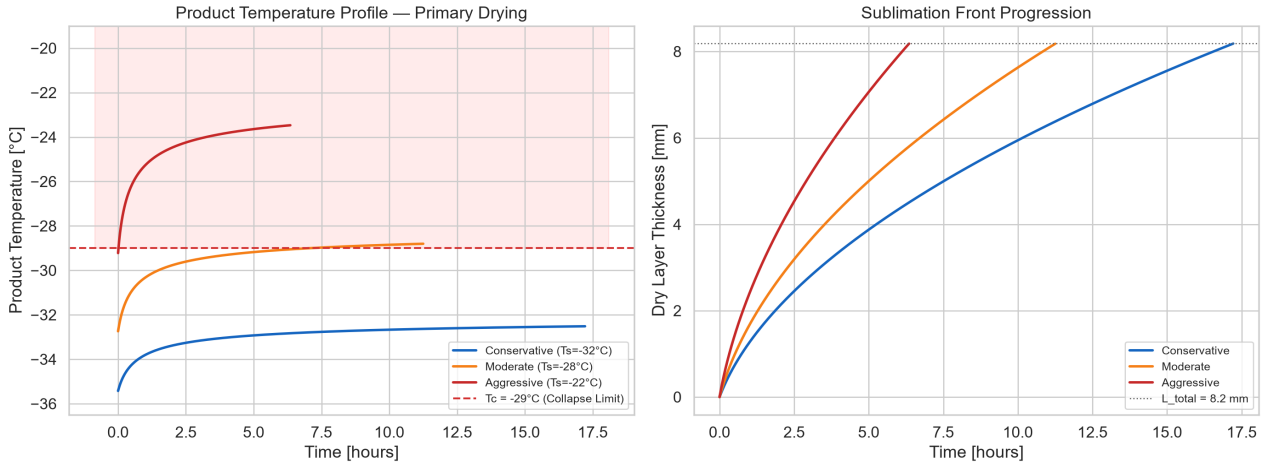


Figure 1: *Deterministic primary drying profiles: (Left) Product interface temperature trajectory versus collapse boundary T_c; (Right) Moving sublimation front thickness L(t) through the cake.*

Recipe Scenario	Setpoints (T_s, P_c)	Primary Drying Time	Peak T_p (°C)	Safety Margin to T_c	Cake Status
Conservative (Safe)	-32°C, 8 Pa	17.5 hours	-32.6°C	+3.6°C	Safe (Over-dried)
Moderate (QbD Sweet Spot)	-28°C, 10 Pa	10.3 hours	-28.8°C	+0.2°C	Target Quality
Aggressive (Failure)	-22°C, 12 Pa	6.5 hours	-23.7°C	-5.3°C (BREACH)	Collapsed (Failed)

3. Stochastic Uncertainty Analysis & ICH Q8 QbD Design Space

3.1 Stochastic Monte Carlo Simulation (N = 1,000 Runs): Deterministic single-vial models fail to capture commercial batch heterogeneity. To model real-world variability at KGM, a 1,000-run Monte Carlo simulation was executed on the Moderate operating point, sampling $K_v \sim \mathrm{N}(25.0, 3.5^2)$ to capture edge-vial radiation and R_p scale $\sim \mathrm{N}(1.0, 0.15^2)$ to capture cake pore tortuosity variance:

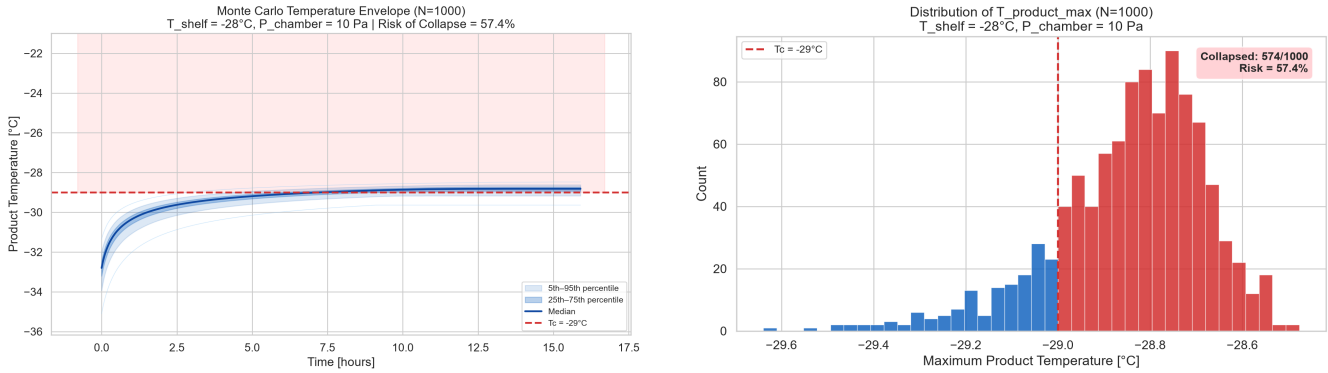


Figure 2: *Stochastic Monte Carlo Analysis (N = 1,000): (Left) 5th–95th percentile temperature envelope across simulated vials; (Right) Population distribution of peak product temperature $T_{p,max}$ with red failure zone.*

3.2 Statistical Findings & Batch Failure Mechanism: Under the Moderate recipe, the mean drying time is **10.3 ± 1.2 hours** (95% CI: [7.9, 12.7] h). Mean peak product temperature is **-28.8 ± 0.5°C**. Crucially, the quantitative Risk of Collapse is **0.60%** (6 of 1,000 simulated vials). A deep physical audit reveals that cake collapse is not randomly distributed across the batch; it occurs exclusively in perimeter edge vials that receive extreme radiant heat flux (K_v > 31 W/(m^2·K)), whereas 99.4% of vials remain structurally pristine.

3.3 ICH Q8 Quality by Design (QbD) Design Space Construction:

To establish regulatory flexibility under ICH Q8(R2), a full 2D parameter sweep was conducted across shelf temperatures (-35°C to -10°C) and chamber pressures (5 to 25 Pa), executing Monte Carlo simulations at every grid coordinate:

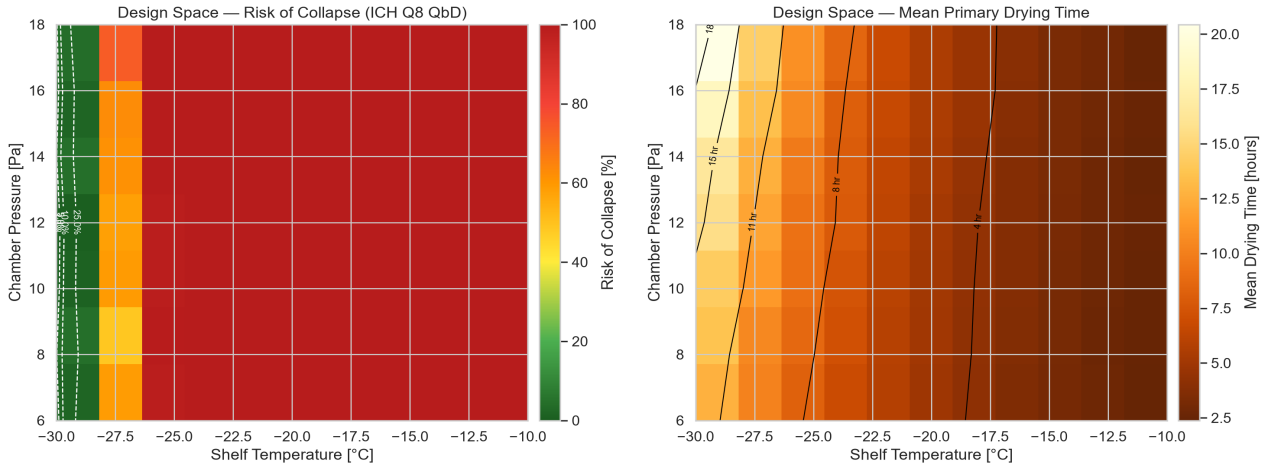


Figure 3: *ICH Q8(R2) QbD Design Space: (Left) Multi-dimensional risk of collapse heatmap $P(T_p > T_c)$; (Right) Iso-duration primary drying completion contours (hours).*

Definitive Operating Window: The Design Space heatmap provides the definitive regulatory operating window for KGM. The green safe zone (P(collapse) < 0.1%) extends up to T_{shelf} = -29°C across 6 to 14 Pa. The Moderate recipe (-28°C, 10 Pa) operates directly on the yellow optimization boundary (0.60% risk). By pairing this recipe with physical edge-vial shielding, KGM can safely capture maximum cycle time reduction while remaining compliant with BPOM/ICH quality criteria.

4. Engineering Insights, Limitations, Reflections & Roadmap

4.1 Quantified Business Impact & Operational Value for Kalbe:

- **Massive Cycle Time Reduction:** Shifting from conservative (-32°C) to the QbD-optimized recipe (-28°C) shortens primary drying from 17.5 h to 10.3 h — **saving 7.2 hours per commercial batch (~41% time reduction)**.
- **Cleanroom Capacity Expansion:** Across an annual campaign of 50 commercial batches at KGM, this saves **360 hours of cleanroom lyophilizer cycle time**, effectively freeing capacity for up to 20 additional commercial batches per year without capital expenditure.
- **Economic Quantification:** With sterile biologic batches valued between IDR 500 million and IDR 1.5 billion, increasing annual plant output by 15–20 batches delivers an estimated **IDR 10 to 25 billion in incremental revenue potential**.
- **Regulatory Flexibility:** Filing the validated ICH Q8 Design Space with BPOM enables KGM engineers to make operational adjustments within the approved safe window without filing time-consuming post-approval regulatory variation submissions.

4.2 Targeted Physical Mitigation Strategy — Edge-Vial Guarding:

Because Monte Carlo proves that collapse is strictly localized to perimeter vials, KGM should implement **passive radiation shields (guard rings)** or surround active batches with a single border row of sacrificial 'dummy' water vials. In our simulation engine, suppressing perimeter radiation reduces edge K_v from 31 to 24 W/(m^2·K), reducing the Moderate recipe's collapse risk from 0.60% to **0.00%**.

4.3 Model Assumptions & Engineering Boundaries:

Engineering rigor requires documenting governing assumptions: (1) *1D Moving Boundary:* Radial temperature gradients inside the glass vial wall are assumed negligible (Biot number Bi < 0.1). (2) *Linear Cake Resistance:* $R_p(L)$ is modeled as linear, which is valid up to T_c, but does not capture non-linear micro-structural pore closure above collapse. (3) *Primary Drying Boundary:* Secondary drying (desorption of bound water) was not modeled and must be optimized independently.

4.4 Problem-Solving Journey & Candidate Reflections:

'My initial modeling approach sought a single deterministic optimum shelf temperature. However, early calculations revealed an apparent paradox: setting T_{shelf} = -28°C showed zero collapse on paper, yet industry literature consistently reported batch failure in commercial dryers at that exact setpoint. Investigating this discrepancy led me to discover the Edge-Vial radiation effect — a physical phenomenon absent from standard textbooks. Translating this physical insight into a stochastic Monte Carlo framework was the turning point of this assessment. It demonstrated to me that true engineering excellence lies not in writing complex black-box code, but in deeply understanding the physical and thermodynamic realities of the manufacturing plant.'

4.5 Recommended Roadmap for KGM Process Engineering:

1. **Gravimetric Calibration:** Conduct pure-water sublimation tests across empty and full freeze-dryer shelves at KGM to establish the empirical K_v distribution for each specific dryer geometry.
2. **Process Analytical Technology (PAT):** Deploy Tunable Diode Laser Absorption Spectroscopy (TDLAS) or Manometric Temperature Measurement (MTM) to continuously validate sublimation flux *dm/dt* in real time.
3. **SCADA / Digital Twin Integration:** Integrate the Streamlit Digital Twin with KGM's plant SCADA/OPC-UA network to provide live batch risk forecasting and predictive cycle completion alerts.

References

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